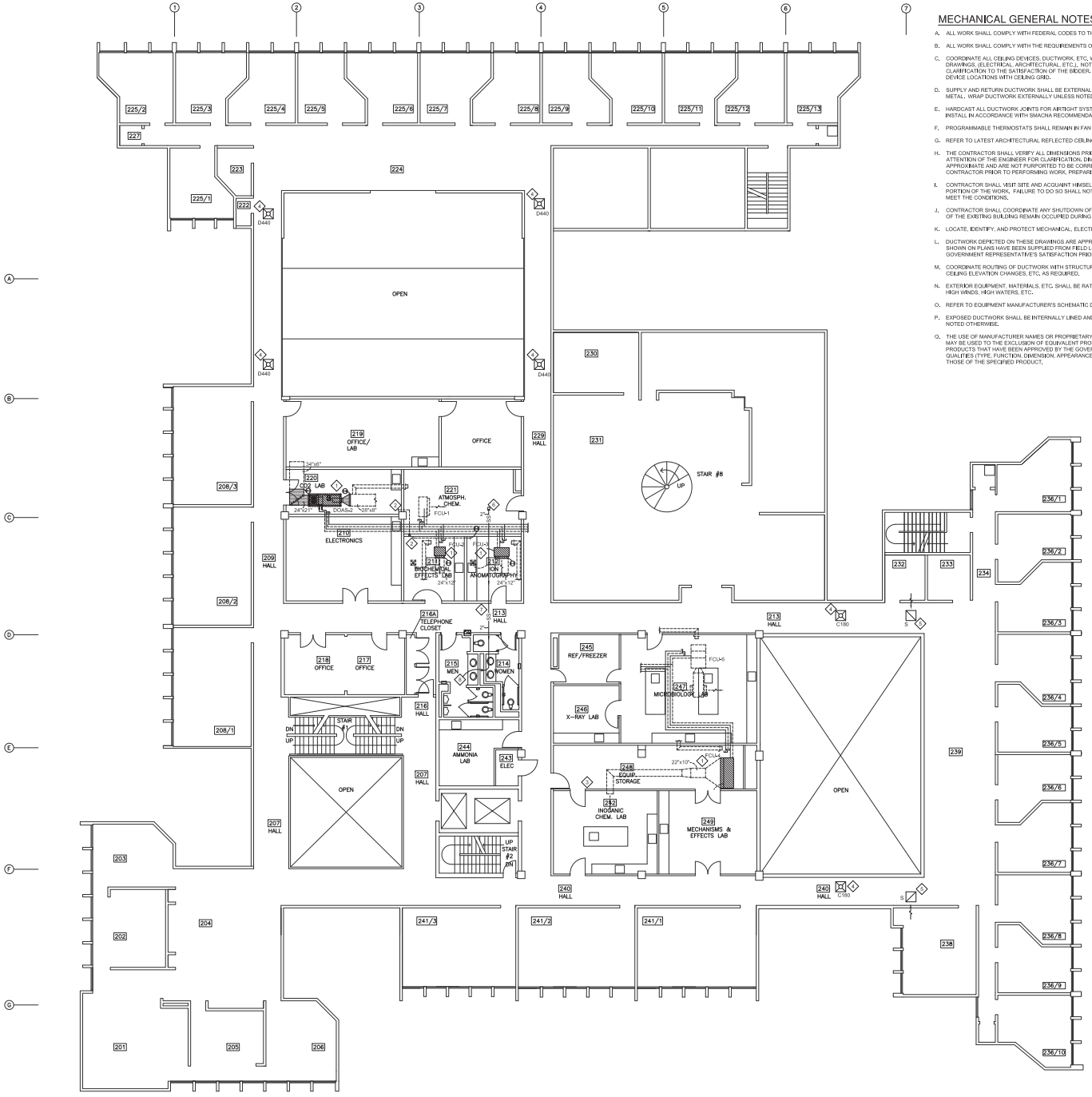


MECHANICAL GENERAL NOTES:

A. ALL WORK SHALL COMPLY WITH FEDERAL CODES TO THE SATISFACTION OF GOVERNMENT.

B. ALL WORK SHALL COMPLY WITH THE REQUIREMENTS OF NFPA 90A & 101, THE CLEAN AIR ACT AND THE AMERICANS WITH DISABILITIES ACT.

C. COORDINATE ALL CEILING DEVICES, DUCTWORK, ETC. WITH LIGHTING, STRUCTURE, ETC., THROUGH GENERAL CONTRACTOR. REFER TO ALL DRAWINGS, (ELECTRICAL, ARCHITECTURAL, ETC.). NOTIFY ARCHITECT/ENGINEER CONCERNING ANY CONFLICTS NOTED PRIOR TO BIDS FOR CLARIFICATION TO THE SATISFACTION OF THE BIDDER. REFER TO LATEST ARCHITECTURAL REFLECTED CEILING PLAN. COORDINATE ALL CEILING DEVICE LOCATIONS WITH CEILING GRID.

D. SUPPLY AND RETURN DUCTWORK SHALL BE EXTERNALLY INSULATED UNLESS NOTED OTHERWISE. DUCT SIZES SHOWN ARE CLEAR METAL TO METAL. WRAP DUCTWORK EXTERNALLY UNLESS NOTED OTHERWISE.

E. HARDCAST ALL DUCTWORK JOINTS FOR AIRTIGHT SYSTEM. REFER TO DETAILS FOR DUCTWORK TAP, SPLITTER, BRANCH CONFIGURATION, ETC. INSTALL IN ACCORDANCE WITH SMACNA RECOMMENDATIONS AND INSTALLATION GUIDELINES.

F. PROGRAMMABLE THERMOSTATS SHALL REMAIN IN FAN "AUTO" MODE.

G. REFER TO LATEST ARCHITECTURAL REFLECTED CEILING PLAN FOR DIFFUSER/GRILLE ACTUAL LOCATIONS.

H. THE CONTRACTOR SHALL VERIFY ALL DIMENSIONS PRIOR TO COMMENCING CONSTRUCTION. ANY DISCREPANCIES SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER FOR CLARIFICATION. DIMENSIONS AND CONDITIONS TYING INTO OR GOVERNED BY EXISTING CONSTRUCTION ARE APPROXIMATE AND ARE NOT PURPORTED TO BE CORRECT. ALL SUCH DIMENSIONS AND CONDITIONS SHALL BE FIELD VERIFIED BY THE CONTRACTOR PRIOR TO PERFORMING WORK, PREPARING SHOP DRAWINGS, OR ORDERING MATERIALS.

I. CONTRACTOR SHALL VISIT SITE AND ACQUAINT HIMSELF THOROUGHLY WITH ALL EXISTING FACILITIES AND CONDITIONS WHICH WOULD AFFECT HIS PORTION OF THE WORK. FAILURE TO DO SO SHALL NOT RELIEVE THE CONTRACTOR FROM THE RESPONSIBILITY OF INSTALLING HIS WORK TO MEET THE CONDITIONS.

J. CONTRACTOR SHALL COORDINATE ANY SHUTDOWN OF SERVICES WITH THE GOVERNMENT REPRESENTATIVE. IT IS THE INTENT THAT PORTIONS OF THE EXISTING BUILDING REMAIN OCCUPIED DURING CONSTRUCTION.

K. LOCATE, IDENTIFY, AND PROTECT MECHANICAL, ELECTRICAL, AND ALL COMMUNICATION SERVICES PASSING THROUGH ENTIRE WORKING AREA.

L. DUCTWORK DEPICTED ON THESE DRAWINGS ARE APPROXIMATE IN NATURE, AND SHOULD ONLY BE USED AS A REFERENCE. ALL INFORMATION SHOWN ON PLANS HAVE BEEN SUPPLIED FROM FIELD LOCATION AND RECORD DRAWINGS. VERIFY EXACT LOCATION AND REQUIREMENTS TO GOVERNMENT REPRESENTATIVE'S SATISFACTION PRIOR TO FURNISHING PROPOSAL.

M. COORDINATE ROUTING OF DUCTWORK WITH STRUCTURAL BEAMS, JOIST, PIPING, CONDUIT, ETC. IN FIELD. PROVIDE OFFSETS AS REQUIRED FOR CEILING ELEVATION CHANGES, ETC. AS REQUIRED.

N. EXTERIOR EQUIPMENT, MATERIALS, ETC. SHALL BE RATED FOR COASTAL ENVIRONMENTS AND SHALL BE PROTECTED FROM SALTY ATMOSPHERE, HIGH WINDS, HIGH WATERS, ETC.

O. REFER TO EQUIPMENT MANUFACTURER'S SCHEMATIC DRAWINGS PRIOR TO ORDERING MATERIALS.

P. EXPOSED DUCTWORK SHALL BE INTERNALLY LINED AND PAINT GRIPPED AND PAINTED (COLOR BY GOVERNMENT REPRESENTATIVE) UNLESS NOTED OTHERWISE.

Q. THE USE OF MANUFACTURER NAMES OR PROPRIETARY PRODUCT NAMES SHALL NOT BE CONSTRUED TO IMPLY THAT ONLY THE NAMED PRODUCT MAY BE USED TO THE EXCLUSION OF EQUIVALENT PRODUCTS BY OTHER MANUFACTURERS. EQUIVALENT PRODUCTS SHALL BE DEFINED AS PRODUCTS THAT HAVE BEEN APPROVED BY THE GOVERNMENT THROUGH THE SUBMITTAL PROCESS AND HAVE BEEN DETERMINED TO HAVE THE QUALITIES (TYPE, FUNCTION, DIMENSION, APPEARANCE, IN-SERVICE PERFORMANCE) AND PHYSICAL PROPERTIES THAT ARE EQUAL TO OR EXCEED THOSE OF THE SPECIFIED PRODUCT.

MECHANICAL KEYNOTES:

1. PROVIDE NEW SUSPENDED AIR HANDLING UNIT IN THIS VICINITY. HANG NEW AIR HANDLER UNIT FROM STRUCTURE IN AN EMERGENCY DRAIN PAN. EMERGENCY DRAIN PAN SHALL EXTEND 6" ALL AROUND PERIMETER OF UNIT WITH A 3/4" INSULATED COPPER DRAIN PIPE WITH BALL VALVE (NC).
2. EXTEND NEW CONDENSATE AND EMERGENCY DRAIN LINE FROM DOAS-2, FC-2, AND FC-3 TO EXISTING OVERHEAD DRAIN LINE IN THIS VICINITY.
3. EXTEND NEW CONDENSATE AND EMERGENCY DRAIN LINE FROM FC-4 TO EXISTING CONDENSATE DROP IN THIS VICINITY.
4. INSTALL NEW SUPPLY DIFFUSER IN SUSPENDED CEILING IN THIS VICINITY, UTILIZE FLEX DUCT FROM EXISTING TAP IN OVERHEAD MAIN TO SUPPLY DIFFUSER.
5. INSTALL NEW RETURN GRILLE IN SUSPENDED CEILING IN THIS VICINITY. ORIENT GRILLE BLADES TO LIMIT VISIBILITY THROUGH GRILLE FROM BELOW.
6. NEW 2 INCH SANITARY SEWER DROPS DOWN FROM 3RD FLOOR BREAKROOM SINK IN THIS VICINITY.
7. ROUTE NEW SANITARY SEWER LINE APPROXIMATELY AS SHOWN.
8. CONNECT TO EXISTING 4 INCH VERTICAL SANITARY SEWER RISER IN PLUMBING CHASE IN THIS VICINITY. ALL PENETRATIONS INTO CHASE SHALL BE MADE ABOVE EXISTING CEILING. PATCH AND SEAL CHASE WALL AIR TIGHT.

NOTE:
CONTRACTOR SHALL COORDINATE EXISTING AND NEW UNIT TAGS WITH BUILDING MANAGEMENT SYSTEM (BMS) AND MAKE ALL NECESSARY MODIFICATIONS TO BMS USER INTERFACE.

1 Second Floor Mechanical - New
1/8" = 1'-0"

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ADG Project No.
13124-AO

Ahtna Global, LLC

UNITED STATES DEPARTMENT OF COMMERCE NATIONAL OCEANIC & ATMOSPHERIC ADMINISTRATION

HURRICANE REPAIRS AT AOML MIAMI

4301 RICKENBACKER CSWY
MIAMI (VIRGINIA KEY), FLORIDA 33149

Date Issued
July 29, 2019

Drawn By
KAS

Checked By
LWB

File Name
13124-AO M105.dwg

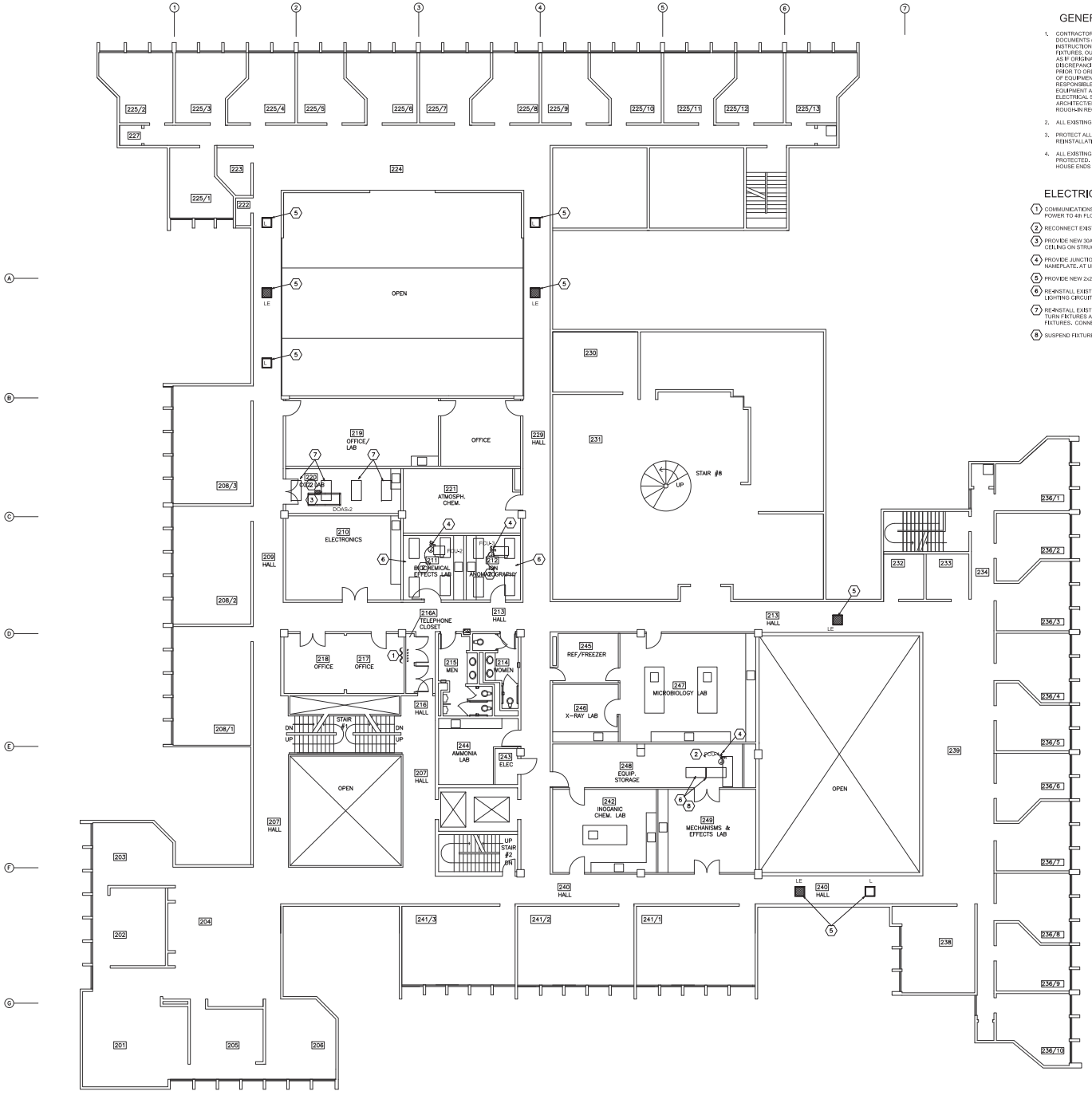
Revision	Date

Drawing Title:
SECOND FLOOR MECHANICAL - NEW

Professional Seal:



Sheet No.
M-105



1 Second Floor - Lighting, Power & HVAC Plan
1/8" = 1'-0"

GENERAL DEMOLITION NOTES:

1. CONTRACTOR SHALL REFER TO ALL OTHER PORTIONS OF THE CONTRACT DOCUMENTS (PLANS, SPECIFICATIONS, ADDENDA, ARCHITECTURAL SUPPLEMENTAL INSTRUCTIONS AND ANY APPROVED CHANGE ORDERS) AND PROVIDE ALL LIGHT FIXTURES, OUTLETS, TELE/DATA OUTLETS, SPEAKERS, AND ASSOCIATED CIRCUITRY AS IF ORIGINALLY INCLUDED ON THE ELECTRICAL PLANS, IF THERE ARE ANY DISCREPANCIES, CONTRACTOR SHALL NOTIFY ARCHITECT/ENGINEER IN WRITING PRIOR TO ORDERING EQUIPMENT, ROUGH-IN FOR EQUIPMENT AND/OR INSTALLATION OF EQUIPMENT. PRIOR TO ROUGH-IN OF EQUIPMENT, CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING COPIES OF APPROVED SHOP DRAWINGS OF SUCH EQUIPMENT AND REVIEWING SAID SUBMITTALS TO ENSURE COMPATIBILITY WITH THE ELECTRICAL SYSTEM. CONTRACTOR SHALL IMMEDIATELY NOTIFY ARCHITECT/ENGINEER IN WRITING OF ANY DISCREPANCIES BETWEEN THE REQUIRED ROUGH-IN REQUIREMENTS AND THE ELECTRICAL SYSTEM.
2. ALL EXISTING DEVICES SHALL REMAIN UNLESS SPECIFICALLY NOTED OTHERWISE.
3. PROTECT ALL EXISTING DEVICES, EQUIPMENT, ETC. FROM DAMAGE PENDING REINSTALLATION AS INDICATED ON SHEETS.
4. ALL EXISTING FIBER OPTIC CABLE SHALL REMAIN IN AREA AND SHALL BE PROTECTED. CONTRACTOR SHALL PROVIDE 12"x12" JUNCTION BOX (TOTAL OF 5) TO HOUSE ENDS OF EXISTING CABLES.

ELECTRICAL KEYNOTES:

1. COMMUNICATIONS & POWER CONDUITS FROM BELOW TO ABOVE. CONDUITS INCLUDE POWER TO 4th FLOOR & SINGLE MODE FIBERS FROM 1st FLOOR.
2. RECONNECT EXISTING CIRCUIT TO NEW UNIT.
3. PROVIDE NEW 30A/3P 600v NEMA 1, FDS FUSED AT EQUIPMENT NAMEPLATE ABOVE CEILING ON STRUCTURE ADJACENT TO UNIT AND MAINTAIN PROPER CLEARANCES.
4. PROVIDE JUNCTION BOX WITH MOTOR RATED SWITCH FUSED FOR EQUIPMENT NAMEPLATE. AT UNIT CONCEALED ABOVE CEILING.
5. PROVIDE NEW 2x2 FIXTURE. RECONNECT TO EXISTING LIGHTING CIRCUIT.
6. RE-INSTALL EXISTING LIGHT FIXTURE. CLEAN & RE-LAMP. CONNECT TO EXISTING LIGHTING CIRCUIT.
7. RE-INSTALL EXISTING LIGHT FIXTURES. FIXTURES TO BE SUSPENDED FROM STRUCTURE. TURN FIXTURES AS INDICATED. SPACE EVENLY WITHIN SPACE. CLEAN AND RE-LAMP FIXTURES. CONNECT LIGHTING CIRCUIT.
8. SUSPEND FIXTURE FROM STRUCTURE USING CHAIN HANGERS.

100% Submittal

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ADG Project No.
13124-AO

Ahtna Global, LLC

UNITED STATES DEPARTMENT OF COMMERCE NATIONAL OCEANIC & ATMOSPHERIC ADMINISTRATION

HURRICANE REPAIRS AT AOML MIAMI

4301 RICKENBACKER CSWY
MIAMI (VIRGINIA KEY), FLORIDA 33149

Date Issued
July 29, 2019
Drawn By
PMB
Checked By
LAM
File Name
13124-AO E-105.dwg

Revision	Date

Drawing Title:
SECOND FLOOR - LIGHTING, POWER & HVAC PLAN

Sheet No.
E-105

STATE OF LOUISIANA
LARRY A. MAYNARD
REG. No. 29187
REGISTERED PROFESSIONAL ENGINEER
ELECTRICAL ENGINEERING



TRANE

MODEL NO: BCHD054G2L0A4G04Z000000A01F1200000C00000
SERIAL NO: H21C17949
SALES ORDER NO: V8A328AB
RATED VOLTAGE: 460 HZ: 60 PH: 3
MINIMUM CIRCUIT AMPACITY: 3.1 AMPS
MAXIMUM OVERCURRENT PROTECTION: 15 AMPS
SHORT–CIRCUIT CURRENT: 5 KA RMS SYMMETRICAL, 460V MAXIMUM
BLOWER MOTOR: 2.5 FLA 1.0 HP
FACTORY PROGRAMMED RPM: 818
MAXIMUM ALLOWABLE FAN RPM: 1500

DO NOT EXCEED 130F (55C) LEAVING AIR TEMPERATURE OFF THE DRAW–THRU COILS

IMPORTANT: UNIT COILS ARE DESIGNED FOR WATER ONLY – DO NOT USE STE

MAXIMUM INLET WATER TEMPERATURE: 200F (93C)
MAXIMUM WATER PRESSURE: 300 PSI (20.7 BAR)

Made in U.S.A.


2021-03-02
NEMA2
TRANE
Enclosure Type 2
Class 2 Supply
Assembled in USA
X13612093005
c US
4011
11 in-lb / 1.4 Nm
AC/DC 24 V 50/60 Hz
0.6 VA 0.3 W
M 75 s
WH RD BK
3 2 1

AIR HANDLING UNIT SCHEDULE

UNIT No.	AREA SERVED	SUPPLY AIR CFM	OUTSIDE AIR CFM	EXTERNAL S.P. - IN/WTR	FAN MOTOR HP	ELECTRIC SERVICE	AIR BY ZONES ZONE 1	ZONE 2	ZONE 3	ZONE 4	ZONE 5	ZONE 6	ZONE 7	ZONE 8	ZONE 9	TOTAL CAPACITY BTUH/HR	CHILLED WATER IN (°F)	CHILLED WATER OUT (°F)	CHILLED WATER GPM	CHILLED WATER PD (FT)	EAT (°F) DB	EAT (°F) WB	LAT (°F) DB	LAT (°F) WB	MAX. PRESS. DROP FT/WTR	MAX. FACE VEL. FPM	REMARKS	NOTES
AHU-1	1ST FLOOR	18,000	1,540	.30	10	480/3/60	4,610	6,440	1,530	2,080	2,000	1,340	---	---	---	488,610	45	55	98		76.38	63.0	55.0	53.6	10	550	* EXISTING UNIT TO REMAIN	
AHU-2	SHOP	2,720	190	.17	2	480/3/60	---	---	---	---	---	---	---	---	---	70,180	45	55	14		76.12	62.6	55.0	53.6	10	550	* EXISTING UNIT TO REMAIN	
AHU-3																											* EXISTING UNIT TO BE REMOVED - SPACE TO BE FED BY AHU-1	
AHU-5	2ND FLOOR	31,900	2,600	1.8	30	480/3/60	9,660	1,180	2,330	2,630	2,200	6,240	3,430	1,850	2,180	890,000	45	55	173		76.6	64.2	55.0	54.5	5.9	520	YORK SOLUTION XT1-108x108 OR APPROVED EQUAL	1, 2, 3, 4, 5, 6
AHU-6	3RD FLOOR (PARTIAL)	25,000	2,000	1.8	25	480/3/60	---	---	2,200	1,240	2,920	8,940	7,290	5,750	---	689,000	45	52	195		77.2	64.8	55.0	54.0	3.8	480	YORK SOLUTION XT1-102x96 OR APPROVED EQUAL	1, 2, 3, 4, 5, 6
AHU-7	4TH FLOOR	3,040	440	1.8	2	480/3/60	900	2,140	---	---	---	---	---	---	---	98,000	45	56	17		77.8	65.4	55.0	54.0	8.1	465	YORK SOLUTION XT1-33x54 OR APPROVED EQUAL	1, 2, 3, 4, 5, 6
AHU-8	5TH FLOOR	8,860	450	1.8	7.5	480/3/60	2,630	3,760	2,470	---	---	---	---	---	---	288,000	45	55	57		76.0	63.6	55.0	54.0	4.5	475	YORK SOLUTION XT1-60x66 OR APPROVED EQUAL	1, 2, 3, 4, 5, 6
AHU-11	3RD FLOOR (PARTIAL)	25,000	2,000	1.8	25	480/3/60	---	---	---	---	---	---	---	---	---	723,000	45	55	145		76.6	64.2	55.0	54.0	8.2	499	YORK SOLUTION XT1-102x96 OR APPROVED EQUAL	1, 2, 3, 4, 5, 6

NOTES:
1. RTU SHALL BE INTERNALLY FUSED FOR SINGLE POINT CONNECTION.
2. UNITS SHALL BE VERTICAL CONFIGURATION DOUBLE WALL AHUS.
3. UNITS SHALL BE HORIZONTAL CONFIGURATION DOUBLE WALL AHUS. FIELD TAP SUPPLY DUCT CONNECTIONS IN SUPPLY FAN SECTION.
4. COOLING COILS SHALL HAVE A MINIMUM OF 8 ROWS AND 10 FINS PER INCH.
5. HEATING COILS (PRE-HEAT AND RE-HEAT) SHALL HAVE A MINIMUM OF 2 ROWS AND 10 FINS PER INCH.
6. CONTROLLER TO INTERFACE TO EXISTING FACILITY BUILDING AUTOMATION SYSTEM.

FAN COIL UNIT SCHEDULE

UNIT No.	TOTAL COOLING CAP. BTUH/HR	TOTAL SENSIBLE LOAD BTUH/HR	WATER FLOW GPM	EAT (°F) DB	EAT (°F) WB	TOTAL SUPPLY CFM	OUTSIDE AIR CFM	MOTOR WATTS MAX. / HP	E.S.P.	ELECTRIC SERVICE	AREA SERVED	REMARKS	NOTES
FCU-1											MASS PROPERTIES LAB.	*EXISTING UNIT TO REMAIN	
FCU-2	5,660	4,990	1.2	75.0	62.6	280	50	180 W	0.50"	120/1/60	GEO. CHRON. LAB	ENVIRO-TEC HLP2S OR APPROVED EQUAL	
FCU-3	8,980	8,310	1.8	75.0	62.6	380	50	180 W	0.50"	120/1/60	SPECIAL GEO. CHRON. LAB	ENVIRO-TEC HLP40 OR APPROVED EQUAL	
FCU-4	26,070	25,400	5.2	75.0	62.6	1180	100	180 W	0.50"	120/1/60	PHY. OCEAN LAB.	ENVIRO-TEC HPP12 OR APPROVED EQUAL	
DOAS-2	86,000	40,000	17.0	91.0	79.0	1,110	1,110	1 HP	0.50"	480/3/60	PRIMARY AIR	YORK SOLUTION XT1-27x30 OR APPROVED EQUAL	
FCU-6											CHEM. OCEAN	*EXISTING UNIT TO REMAIN	
FCU-7											MECHANICAL ROOM 406	*EXISTING UNIT TO REMAIN	
FCU-8	88,000	74,000	17.0	76.6	64.2	3,170	260	2 HP	0.50"	480/3/60	HYDROCARBON LAB	YORK SOLUTION XT1-39x45 OR APPROVED EQUAL	
FCU-G1											GROUND FLOOR	*EXISTING UNIT TO REMAIN	
FCU-M2											MECHANICAL ROOM 407	*EXISTING UNIT TO REMAIN	

NOTES:
1. FAN COIL UNIT SHALL HAVE MINIMUM 2 ROW COOLING COILS AND HOT WATER COILS.
2. FAN COIL SHALL BE HORIZONTAL RECESSED WITH BOTTOM STAMPED RETURN LOUVER AND FRONT DUCT COLLAR OUTLET. ALL FAN COILS SHALL HAVE FACTORY SUPPLIED VALVE PACKS.
3. PROVIDE CONDENSATE PUMP (LITTLE GIANT VCMA-15 SERIES OR APPROVED EQUAL).
4. FAN COIL SHALL BE LOW PROFILE WITH HORIZONTAL CABINET (FRONT/BACK STAMPED RETURN AND STAMPED SUPPLY GRILLE).
5. CONTROLLER TO INTERFACE TO EXISTING FACILITY BUILDING AUTOMATION SYSTEM.

SPLIT SYSTEM INDOOR UNITS (VRF)

UNIT No.	CONDENSING UNIT	SERVICE	C.F.M. MIN	COOLING CAPACITY	HEATING CAPACITY	EVAP. TEMP. (°F)	AMB. TEMP. (°F)	ENT. TEMP. D.B. °F	ENT. TEMP. W.B. °F	LEAV. TEMP. D.B. °F	LEAV. TEMP. W.B. °F	MCA	MOCP	ELECTRIC SERVICE	REMARKS	NOTES
AC-1	CU-1	MECH. ROOM 406	710	30,000 BTUH	34,000 BTUH	40	95	80	67	55.0	54.0	0.54	15	208/1/60	MITSUBISHI PKFY-P30NKMU-E OR APPROVED EQUAL	1, 2, 3, 4
AC-2	CU-1	MECH. ROOM 406	710	30,000 BTUH	34,000 BTUH	40	95	80	67	55.0	54.0	0.54	15	208/1/60	MITSUBISHI PKFY-P30NKMU-E OR APPROVED EQUAL	1, 2, 3, 4

NOTES & OPTIONS:
1. AIR HANDLING UNIT SHALL BE INTERNALLY FUSED AND WIRED FOR SINGLE POINT CONNECTION.
2. ALL UNITS SHALL BE SET UP TO CYCLE WHEN SATISFIED.
3. UNIT SHALL BE EQUIPPED WITH CONDENSATE PUMP. PUMP SHALL BE INTERLOCKED WITH AHU TO SHUT DOWN UNIT DURING PUMP FAILURE. COORDINATE DRAIN LOCATION WITH PLUMBING CONTRACTOR.
4. UNIT SHALL BE CONNECTED TO A PROGRAMMABLE THERMOSTAT AS WELL AS INTERLOCKED WITH BUILDING MANAGEMENT SYSTEM.

SPLIT SYSTEM OUTDOOR UNITS (VRF)

UNIT No	SERVICE	BTUH COOLING OUTPUT	EVAP TEMP	AMBIENT TEMP	MCA	MOCP	ELECTRIC SERVICE	MIN E.E.R.	REMARKS	NOTES
CU-1	AC-1, AC-2	60,000	40° F	95°F	36	45	208/1/60	11.1	MITSUBISHI PUMY-P60NKMU2-BS OR APPROVED EQUAL	1, 2, 3, 4

NOTES & OPTIONS:
1. SHALL BE PROVIDED WITH LOW AMBIENT CONTROLS.
2. SHALL BE PROVIDED WITH WIND BAFFLE ACCESSORY.
3. CONDENSER COILS SHALL BE COATED WITH PHENOLIC (COASTAL/SALT AIR) COATING SYSTEM.
4. INTERLOCK UNIT WITH EXISTING FACILITY BUILDING AUTOMATION SYSTEM.

NOTE:

ALL THREE PHASE EQUIPMENT SHALL BE EQUIPPED WITH PHASE MONITORS TO PROTECT AGAINST PHASE LOSS, UNBALANCE, OR REVERSAL WITH ALARM INTERFACE TO BUILDING MANAGEMENT SYSTEM AND AUTO RESTART.

CONTRACTOR SHALL VERIFY THAT, UPON COMPLETION OF WORK, ALL EXISTING AND NEW UNITS ARE INTERLOCKED WITH THE BUILDING MANAGEMENT SYSTEM AND MATCH THE NEW/REVISED UNIT TAGS INDICATED WITHIN THESE DOCUMENTS.

SPLIT SYSTEM INDOOR UNITS (CRAC)

UNIT No.	CONDENSING UNIT	SERVICE	C.F.M. MIN	TOTAL COOLING CAPACITY	SENSIBLE COOLING CAPACITY	EVAP. TEMP. (°F)	AMB. TEMP. (°F)	ENT. TEMP. D.B. °F	ENT. TEMP. W.B. °F	LEAV. TEMP. D.B. °F	LEAV. TEMP. W.B. °F	MOTOR PWR & QTY.	E.S.P.	TYPE	QTY.	SIZE (TON)	MCA	MOCP	ELECTRIC SERVICE	REMARKS	NOTES
CRAC-1	CU-CRAC-1	DATA CENTER - 3RD FL	20,000	481,600 BTUH	453,200 BTUH	40	95	75	61.1	54.1	52.5	4 @ 3.6 KW	1 INCH	SCROLL	1 INCH	20 (EA.)	95.0	130	460/3/60	COMPU-AIRE SYSTEM 2100 - CAA-4034 OR APPROVED EQUAL	1, 2, 3, 5

NOTES & OPTIONS:
1. AIR HANDLING UNIT SHALL BE INTERNALLY FUSED AND WIRED FOR SINGLE POINT CONNECTION.
2. ALL UNITS SHALL BE SET UP TO CYCLE WHEN SATISFIED.
3. UNIT SHALL BE EQUIPPED WITH CONDENSATE PUMP. PUMP SHALL BE INTERLOCKED WITH AHU TO SHUT DOWN UNIT DURING PUMP FAILURE. COORDINATE DRAIN LOCATION WITH PLUMBING CONTRACTOR.
4. UNIT SHALL BE CONNECTED TO A TEMPERATURE SENSOR WITHIN THE DATA CENTER AS WELL AS INTERLOCKED WITH BUILDING MANAGEMENT SYSTEM.
5. UNIT SHALL BE INTERLOCKED WITH BUILDING MANAGEMENT SYSTEM.

SPLIT SYSTEM OUTDOOR UNITS (CRAC)

UNIT No	SERVICE	BTUH COOLING OUTPUT	EVAP TEMP	AMBIENT TEMP	MCA	MOCP	ELECTRIC SERVICE	MIN E.E.R.	REFRIGERANT CIRCUITS	MOTOR HP & QTY.	REMARKS	NOTES
CU-CRAC-1	DATA CENTER - 3RD FL	481,600	40° F	95°F	8.0	15	460/3/60	--	2	4 @ .33 HP	COMPU-AIRE KCM042-T4A-MV OR APPROVED EQUAL	1, 2, 3, 4

NOTES & OPTIONS:
1. SHALL BE PROVIDED WITH LOW AMBIENT CONTROLS.
2. SHALL BE PROVIDED WITH WIND BAFFLE ACCESSORY.
3. CONDENSER COILS SHALL BE COATED WITH PHENOLIC (COASTAL/SALT AIR) COATING SYSTEM.
4. INTERLOCK UNIT WITH EXISTING FACILITY BUILDING AUTOMATION SYSTEM.

FAN SCHEDULE

No	SERVICE	MIN. CFM	EXT. SP	RPM	MAX SONES	FAN POWER	TYPE	DRIVE	ELECTRIC SERVICE	REMARKS	NOTES
EF-15	URAA AREA PLOTTER EXHAUST	100	.50"	1049	2.5	41.7W	CENT.	DIRECT	120/1/60	COOK MODEL GN-148 OR APPROVED EQUAL	1, 2

NOTES:
1. FAN SHALL BE INTERLOCKED WITH BUILDING MANAGEMENT SYSTEM. FAN SHALL OPERATION SHALL BE CONTROLLED BY WALL SWITCH WITH OVERRIDE AND SCHEDULING CAPABILITIES FROM BMS.
2. FAN SHALL BE EQUIPPED WITH SOLID STATE SPEED CONTROL AND SPRING (OR RUBBER) ISOLATORS FOR SUSPENDED INSTALLATION APPLICATION. PROVIDE ALL NECESSARY ACCESSORIES, RELAYS, WIRING, ETC. FOR A COMPLETE AND WORKING SYSTEM MEETING REQUIREMENTS OF DRAWINGS AND APPLICABLE SPECIFICATION SECTIONS.

VARIABLE AIR VOLUME BOX SCHEDULE

UNIT No.	AREA SERVED	COOLING CFM MIN	COOLING CFM MAX	INLET SIZE	MAX SP LOSS ACROSS BOX	HEATING CAPACITY BTUH	HEATING CFM	HEATING AIR TEMP ENT	HEATING AIR TEMP LEAV	HEATING KW STRIP KW	HEATING KW STRIP STAGES	SERVICE	REMARKS	NOTES
VAV 36	311/B, 311/C, 311/D	200	1000	10	.25"	8,530	200	55.0	94.5	2.5	3	480/3/60	TITUS MODEL DESV OR APPROVED EQUAL	1, 2, 3
VAV 37	312/A, 310J, CORRIDOR	200	830	8	.25"	8,530	200	55.0	94.5	2.5	3	480/3/60	TITUS MODEL DESV OR APPROVED EQUAL	1, 2, 3
VAV 38	328A, LIBRARY UPPER LEVEL	240	940	10	.25"	13,660	320	55.0	94.5	4.0	3	480/3/60	TITUS MODEL DESV OR APPROVED EQUAL	1, 2, 3
VAV 39	310/G, 310/H	190	760	8	.25"	8,530	200	55.0	94.5	2.5	3	480/3/60	TITUS MODEL DESV OR APPROVED EQUAL	1, 2, 3
VAV 40	311/A, 311/E, 311/F	200	840	8	.25"	8,530	200	55.0	94.5	2.5	3	480/3/60	TITUS MODEL DESV OR APPROVED EQUAL	1, 2, 3
VAV 41	311	155	610	8	.25"	8,530	200	55.0	94.5	2.5	3	480/3/60	TITUS MODEL DESV OR APPROVED EQUAL	1, 2, 3
VAV 42	312	240	2270	14	.25"	10,240	240	55.0	94.6	3.0	3	480/3/60	TITUS MODEL DESV OR APPROVED EQUAL	1, 2, 3
VAV 43	328/B	400	4000	16	.25"	10,240	400	55.0	78.8	3.0	3	480/3/60	TITUS MODEL DESV OR APPROVED EQUAL	1, 2, 3
VAV 44	329	150	590	8	.25"	8,530	200	55.0	94.6	2.5	3	480/3/60	TITUS MODEL DESV OR APPROVED EQUAL	1, 2, 3
VAV 45	308/1, 308/2, 308/3, 3rd FL NE CORRIDOR	680	2700	14	.25"	30,710	720	55.0	94.5	9.0	3	480/3/60	TITUS MODEL DESV OR APPROVED EQUAL	1, 2, 3
VAV 46	320, 323/1 - 323/9, 323/11	1250	5000	24"X16"	.25"	68,240	1580	55.0	95.0	20.0	3	480/3/60	TITUS MODEL DESV OR APPROVED EQUAL	1, 2, 3
VAV 47	320, 323/10, 323/12 - 323/18, 325	1250	5000	24"X16"	.25"	68,240	1580	55.0	95.0	20.0	3	480/3/60	TITUS MODEL DESV OR APPROVED EQUAL	1, 2, 3

NOTES:
1. CONTROLS CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING 24 VOLT POWER TO ALL VAV BOXES FOR ACTUATOR AND CONTROLS.
2. VAV BOX SUPPLIER SHALL COORDINATE WITH MECHANICAL CONTRACTOR PRIOR TO ORDERING BOXES TO ORDER CONTROL PANELS ON ACCESSIBLE SIDE OF VAV BOX (AFTER FINAL DUCT WORK LAYOUT HAS BEEN DETERMINED IN FIELD).
3. CONTROLLER TO INTERFACE TO EXISTING FACILITY BUILDING AUTOMATION SYSTEM.

INSERTION TYPE VAV SCHEDULE

UNIT No.	AREA SERVED	SUPPLY CFM MIN	SUPPLY CFM MAX	SERVICE	REMARKS	NOTES
VAV 33	1ST FL ATRIUM	115	440	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 34	1ST FL CONF. RM.	126	470	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 35	112, 113, 114, 117	110	425	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 48	AHU-5	330	2,180	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 49	AHU-5	395	2,630	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 50	AHU-5	1,480	9,860	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 51	AHU-5	280	1,850	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 52	AHU-5	330	2,200	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 53	AHU-5	350	2,330	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 54	AHU-5	940	6,240	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 55	AHU-5	180	1,180	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 56	AHU-5	515	3,430	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 57	AHU-6	885	5,940	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 58	AHU-6	865	5,790	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 59	AHU-6	1095	7,290	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 60	AHU-6	330	2,200	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 61	AHU-6	190	1,240	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 62	AHU-6	440	2,920	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 63	AHU-7	135	900	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 64	AHU-7	325	2,140	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 65	AHU-8	375	2,470	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 66	AHU-8	565	3,760	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5
VAV 67	AHU-8	395	2,630	120/1/60	TITUS MODEL QCV	1, 2, 3, 4, 5

NOTES:
1. CONTROLS CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING 24 VOLT POWER TO ALL KIT ASSEMBLIES FOR ACTUATOR AND CONTROLS.
2. UNIT SUPPLIER SHALL COORDINATE WITH MECHANICAL CONTRACTOR PRIOR TO ORDERING BOXES TO ORDER CONTROL PANELS ON ACCESSIBLE SIDE OF ASSEMBLY (AFTER FINAL DUCT WORK LAYOUT SHALL BE DETERMINED IN FIELD).
3. UNIT SHALL BE EQUIPPED WITH THE FOLLOWING FEATURES:
   24 V CONTROL TRANSFORMER
   DUST TIGHT ENCLOSURE SEAL
   DISCONNECT SWITCH
   ACTUATOR
4. CONTROLLER TO INTERFACE TO EXISTING FACILITY BUILDING AUTOMATION SYSTEM.
5. REFER TO MECHANICAL PLANS AND ENLARGED MECHANICAL ROOM SHEETS FOR DUCT SIZES.

GRILLE AND DIFFUSER SCHEDULE

SYMBOL	SIZE	SERVICE	LOCATION	FINISHED	REMARKS	NOTES
A	12"x12"	SUPPLY	CEILING	BY ARCH	TITUS MODEL TMS-AA, 3 CONE, W/ 6" NECK OR APPROVED EQUAL	1, 2, 3, 4, 5, 6
B	24"X24"	SUPPLY	CEILING	BY ARCH	TITUS MODEL TMS-AA, 3 CONE, W/ 8" NECK OR APPROVED EQUAL	1, 2, 3, 4, 5, 6
C	24"x24"	SUPPLY	CEILING	BY ARCH	TITUS MODEL TMS-AA, 3 CONE, W/ 10" NECK OR APPROVED EQUAL	1, 2, 3, 4, 5, 6
D	24"x24"	SUPPLY	CEILING	BY ARCH	TITUS MODEL TMS-AA, 3 CONE, W/ 12" NECK OR APPROVED EQUAL	1, 2, 3, 4, 5, 6
E	18"x8"	SUPPLY	DUCT	BY ARCH	TITUS MODEL 300FL, ALUMINUM DOUBLE DEFLECTION OR APPROVED EQUAL	3, 5
F	18"x10"	SUPPLY	DUCT	BY ARCH	TITUS MODEL 250-AA, ALUMINUM CURVED BLADE DUCT MOUNT OR APPROVED EQUAL	3, 5
G	42"x14"	SUPPLY	DUCT	BY ARCH	TITUS MODEL 300FL, ALUMINUM DOUBLE DEFLECTION OR APPROVED EQUAL	3, 5
H	48"x48"	SUPPLY	DUCT	BY ARCH	TITUS MODEL 250-AA, ALUMINUM CURVED BLADE DUCT MOUNT OR APPROVED EQUAL	3, 5
R	36"x24"	RETURN	WALL	BY ARCH	TITUS MODEL 350FRL, STEEL SINGLE DEFLECTION OR APPROVED EQUAL	5
S	24"x24"	RETURN	CEILING	BY ARCH	TITUS MODEL 355FS, ALUMINUM SINGLE DEFLECTION OR APPROVED EQUAL	2, 4, 5, 7
W	12"x12"	EXHAUST	DUCT	BY ARCH	TITUS MODEL 50F, EGGCRATE GRILLE W/ REMOVABLE CORE OR APPROVED EQUAL	2, 4, 5
Y	24"x12"	RETURN	CEILING	BY ARCH	TITUS MODEL 50F, EGGCRATE GRILLE W/ REMOVABLE CORE OR APPROVED EQUAL	2, 4, 5
Z	24"x24"	RETURN	CEILING	BY ARCH	TITUS MODEL 50F, EGGCRATE GRILLE W/ REMOVABLE CORE OR APPROVED EQUAL	2, 4, 5
X	--	--	--	--	DENOTES AN EXISTING DIFFUSER/GRILLE WHICH IS TO BE RE-USED	--

NOTES:
1. TOPS OF ROUND NECK DIFFUSERS SHALL BE INSULATED. REFER TO DETAIL.
2. SHEET METAL CONTRACTOR TO PROVIDE SHEET METAL CEILING BOX FOR ATTACHMENT OF RECTANGLE DUCTWORK TO RA/EA AIR REGISTER.
3. ALL SUPPLY DIFFUSERS SHALL BE 4-WAY PATTERN UNLESS SHOWN OTHERWISE ON PLANS OR REQUIRED AS PER DETAIL.
4. VERIFY CEILING TYPE PRIOR TO ORDERING MATERIAL. REFER TO ARCHITECTURAL REFLECTED CEILING PLAN.
5. DIFFUSER/GRILLE SHALL BE FINISHED TO MATCH MOUNTING LOCATION. VERIFY WITH CONTRACTING OFFICER PRIOR TO ORDERING MATERIAL.
6. MOUNT DIFFUSER IN CENTER OF TILE. PROVIDE BLANK OFF PLATES TO FIT FULL TILE SPACE.
7. ORIENT RETURN GRILLE INSTALLATION TO REDUCE VISIBILITY THROUGH GRILLE FROM BELOW.

MECHANICAL LEGEND

SYMBOL	DESCRIPTION
TYP.	TYPICAL
AHU	AIR HANDLING UNIT
BTUH	BRITISH THERMAL UNIT PER HOUR
CWS	CHILLED WATER SUPPLY
CWR	CHILLED WATER RETURN
HWS	HEATING HOT WATER SUPPLY
HWR	HEATING HOT WATER RETURN
EF	EXHAUST FAN
FT	FEET
FLA	FULL LOAD AMPS
FCU	FAN COIL UNIT
MCA	MINIMUM CIRCUIT AMPACITY
PD	PRESSURE DROP
SP	STATIC PRESSURE
FSD	COMBINATION FIRE/ISOLATION DAMPER
	DUCTWORK CAP (SEAL AIRTIGHT)
	EXISTING SUPPLY DIFFUSER
	EXISTING RETURN GRILLE
	MECHANICAL EQUIPMENT
	MANUAL DAMPER
	DOMESTIC COLD WATER SUPPLY
	DOMESTIC HOT WATER SUPPLY
	SANITARY SEWER VENT LINE
-SS-	SANITARY SEWER LINE
-SD-	STORM DRAIN LINE
-C-	CONDENSATE DRAIN LINE
	CHILLED WATER SUPPLY PIPE
	CHILLED WATER RETURN PIPE
	NEW SUPPLY DIFFUSER
	NEW RETURN GRILLE
	INSERTION TYPE VAV RETROFIT KIT (TITUS QCV - BASIS OF DESIGN)
	RECTANGULAR LOW PRESSURE DUCTWORK (EXTERNALLY WRAPPED)
	RECTANGULAR MEDIUM PRESSURE DUCTWORK (DUCTMATE CONNECTIONS) (EXTERNALLY WRAPPED)
	EXISTING TO BE REMOVED
	MECHANICAL DEMO - REFER TO SHEET M-106
	VAV BOX
T	TEMPERATURE SENSOR AND/OR THERMOSTAT
M	MOTORIZED DAMPER (UNLESS NOTED OTHERWISE)

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13124-AO

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UNITED STATES DEPARTMENT OF COMMERCE NATIONAL OCEANIC & ATMOSPHERIC ADMINISTRATION

HURRICANE REPAIRS AT AOML MIAMI

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Date Issued: July 29, 2019
Drawn By: KAS
Checked By: LWB
File Name: 13124-AO M601.dwg

Revision	Date

Drawing Title:
MECHANICAL SCHEDULES

Professional Seal:



Sheet No.
M-601